

1. Project Definition

The Face Recognition Based Attendance System is an automated system that uses computer vision and machine learning techniques to identify individuals through facial features and record their attendance in real time using a camera. The system eliminates the need for manual attendance marking and prevents proxy attendance. A secure smart attendance system that combines QR code scanning, live location verification, and face recognition to ensure that attendance is marked only when the student is physically present in the classroom.

2. Project Objective

The main objectives of the project are:

- To automate the attendance marking process using face recognition
- To reduce manual effort and human errors in attendance management
- To prevent fake or proxy attendance and ensure physical classroom presence
- To ensure fast, accurate, and reliable attendance recording
- To store attendance data securely in digital format
- Improve attendance authenticity and combine location and biometric verification

3. Project Scope

The scope of the project includes:

- Capturing facial images of registered users
- Encoding facial features and storing them securely
- Real-time face recognition using a webcam
- Automatic attendance marking with date and time
- Maintaining attendance records in the database.
- Basic anti-spoofing measures (such as blink detection or reflection detection)

Limitations:

- Requires good lighting conditions
- Performance may reduce with poor camera quality
- Advanced spoof detection (3D depth, infrared) is not included

4. Project Modules

4.1 Face Capture Module

- Captures multiple face images of a user
- Stores images in a structured dataset

4.2 Face Encoding Model

- Anti-Spoofing Module (Basic)
- Saves encoded facial data for recognition

4.3 Face Detection Module

- Detects and recognizes faces in real time
- Matches detected faces with stored encodings

4.4 Face Recognition Module

- Matches detected faces with stored encodings
- Prevents duplicate attendance for the same day

4.5 Attendance Management Module

- Uses blink detection or screen reflection detection
- Prevents attendance using photos or videos

5. Project Basic Requirements

5.1 Hardware Requirements

- Laptop or desktop computer
- Webcam (built-in or external)
- Minimum 4 GB RAM (8 GB recommended)
- Processor: Intel i3 or higher

5.2 Software Requirements

- Operating System: Windows 10 / 11
- Python 3.10
- OpenCV, dlib, NumPy and Pandas
- CMake
- Visual Studio Build Tools (C++)

6. Literature Review

6.1 “Smart Attendance System using Facial Recognition” (IJARCCE 2024, DOI:10.17148/IJARCCE.2024.131207)

<https://ijarcce.com/wp-content/uploads/2024/12/IJARCCE.2024.131207.pdf>

- **Implementation Approach:** They built a real-time attendance system that captures student face images via camera, uses CNN for face detection and recognition, and matches faces against stored templates. The system marks attendance and logs it into a database with timestamps, achieving ~94.7 % accuracy under varied conditions and ~1.2 s per recognition cycle.
- **Tech Stack:** TensorFlow.js, CNN deep learning, Webcam input, IndexedDB/MySQL
- **Dataset Used:** Custom dataset with 5 face images per person, Variations in lighting/pose
- **Comparison with Our System:** Their solution focuses solely on webcam-based facial attendance with CNN and basic storage, lacking QR-based GPS logging and advanced anti-spoofing. They used a very small dataset and browser-side recognition, while our project integrates secure facial encoding, QR GPS capture, robust spoof checks, and scalable DB records.

6.2 “Face Recognition based Attendance Management System (IJERT May-2020)”

https://www.researchgate.net/publication/341876647_Face_Recognition_based_Attendance_Management_System

- **Implementation Approach:** They capture live video from a webcam, detect faces using Haar-Cascade, and recognize them using Local Binary Pattern Histogram (LBPH). Registered student images form the database; detected faces are matched to mark attendance. After recognition, attendance records are updated and emailed to faculty at session end.
- **Tech Stack:** Python, OpenCV(Haar-Cascade + LBPH), Webcam interface, Email Integration.
- **Dataset Used:** Custom face image dataset of enrolled students, Multiple sample images per student under varied orientations/lighting
- **Comparison with Our System:** They rely on basic Haar + LBPH for facial attendance and email summaries, without GPS/QR or anti-spoofing. Their dataset is simple and lacks

security encoding. In contrast, our system adds secure encodings, QR-based GPS verification to reduce proxy, advanced spoof detection (blink/reflection), and structured DB storage for real-time attendance tracking.

6.3 IJERT (June-2020) Face Recognition Based Attendance System

<https://www.ijert.org/research/face-recognition-based-attendance-system-IJERTV9IS060615.pdf>

- **Implementation Approach:** They capture live video from a camera and extract frames using OpenCV, then detect and recognize faces with dlib. Recognized faces are matched against a stored student database and attendance is marked with timestamps in the DB. Their system focuses on real-time face detection and recognition under varied lighting without explicit spoof checks.
- **Tech Stack:** Python(implied), OpenCV (video + image processing), dlib (face detection/recognition)
- **Dataset Used:** Custom dataset of enrolled students' face images, Multiple images per student under different angles/illumination
- **Comparison with Our System:** Their system uses basic webcam-based detection/recognition with dlib and marks attendance in a DB. It lacks QR/GPS to curb proxy, robust anti-spoofing like blink/reflection checks, and secure facial encodings. In contrast, our project adds GPS capture via QR, improved security against spoofing, and structured attendance validation.

6.4 “Facial Recognition Attendance System Using Python and OpenCV” (Quest Journals, D05021829)

<https://www.questjournals.org/jses/papers/Vol5-issue-2/D05021829.pdf>

- **Implementation Approach:** The system captures a class image using a camera, creates a student face dataset, trains a recognizer, and crops faces from the photo for matching with stored images. When a face matches, attendance is marked in an Excel file with current date and time, and the final attendance sheet can be emailed to faculty.
- **Tech Stack:** Python, OpenCV, NumPy, HD camera.
- **Dataset used:** Custom student face dataset, Multiple images per student under varied conditions.

- **Comparison with Our System:** Their system uses simple image capture and matching with Python/OpenCV, saving attendance to Excel and optionally emailing it. It doesn't include QR-based GPS verification, secure facial encoding, database persistence, or anti-spoofing. Our project adds GPS capture via QR, stronger spoof detection, secure encodings, and structured DB attendance storage for real-time tracking.

6.5 “Design and Implementation of Face Recognition Attendance Management System” (ITM Web of Conferences, CCCAR 2022)

https://www.itm-conferences.org/articles/itmconf/pdf/2022/07/itmconf_cccar2022_01024.pdf

- **Implementation Approach:** They built a desktop attendance system in Java with a MySQL backend. On punch-in, the camera captures an image via OpenCV, then a perceptual hash filters candidates and the Rainbow Soft face engine compares faces against stored profiles. Successful matches record attendance and calculate work duration.
- **Tech Stack:** Java, OpenCV, NumPy, HD Camera.
- **Dataset used:** Custom student face dataset, Multiple images per student under varied conditions.
- **Comparison with Our System:** Their system uses simple image capture and matching with Python/OpenCV, saving attendance to Excel and optionally emailing it. It doesn't include QR-based GPS verification, secure facial encoding, database persistence, or anti-spoofing. Our project adds GPS capture via QR, stronger spoof detection, secure encodings, and structured DB attendance storage for real-time tracking.

6.6 eSSL

- **Company Background & Establishment:** eSSL is an Indian biometric security brand founded in 2002, known for time and attendance management systems, access control devices, and workforce management solutions across India. They focus on hardware reliability, nationwide support, and wide adoption in offices, factories, and institutions.
- **Implementation Approach:** eSSL's face recognition attendance products use onboard cameras to capture and match facial biometric templates stored locally. Devices support multiple authentication methods (face, fingerprint, card, PIN) and push attendance logs

via TCP/IP or USB to central servers or HR software. Some models include anti-spoofing detection to block fake images or masks.

- **Tech Stack:** Embedded Linux, Built-in face recognition firmware/algorithms, TCP/IP, USB communication, On-device camera modules.
- **Dataset used:** Proprietary enrolled employee/visitor face templates stored per device, no public dataset specified (custom per installation)
- **Product Specifications:**
 - eSSL SFace900 – Face + fingerprint, robust connectivity.
 - eSSL eFace-990 – 4.3" touch screen.
 - eSSL AI Face Venus – LCD display, anti-spoofing.
 - Screen sizes: Typically 4.3"–5" on advanced models.
- **Market Price:**
 - SFace900: ~₹16,500–₹17,900.
 - AI Face Venus: ~₹21,000.
 - EyeFace-99 (premium): ~₹1,00,000 (higher-end).
 - Basic face devices like MB160: ~₹9,500.
- **Comparison with Our System:** eSSL devices are plug-and-play attendance terminals with built-in recognition and optional anti-spoof features but lack QR-based GPS verification and flexible secure facial encodings. Their data lives on device/ERP integrations, not live secured streams. Our system adds GPS capture via QR, liveness checks (blink/reflection), encrypted encodings, and cloud DB attendance for scalable, proxy-resistant tracking.

6.7 Mantra

- **Company Background & Establishment:** Mantra Softech (Mantratec) is an Indian biometric tech company with deep roots in identity solutions, including Aadhaar-certified fingerprint and iris scanners. They deliver scalable workforce attendance and access control systems nationwide, tailored for enterprises, government programs, and offices. They're known for reliability, customizable integration, and cloud/on-premise options.
- **Implementation Approach:** Mantra's attendance devices capture facial biometric data via onboard RGB/infrared cameras, validate against stored templates, and log attendance in real time via TCP/IP, Wi-Fi, or USB. Many models also support multimode biometrics

(face, fingerprint, RFID/card) and integrate with HRMS/ERP systems. Devices aim for fast (<0.8s) recognition under varied lighting and user conditions.

- **Tech Stack:** On-device face recognition algorithms (firmware), TCP/IP & USB communication, Built-in LCD/TFT display modules.
- **Dataset used:** Proprietary enrolled user face templates stored per device, no public benchmark dataset specified (custom per installation)
- **Product Specifications:**
 - Screen size: ~2.4"–2.8" TFT color display (varies by model)
 - Multimodal authentication: Face, fingerprint, RFID/card
 - Communication: TCP/IP, USB, Wi-Fi (optional)
 - Face capacity: ~1,000–1,500 users (basic models)
- **Market Price:**
 - Basic face device ~₹6,000–₹10,000 (e.g., BioFace-MSD1K)
 - Slightly advanced/expanded models ~₹8,000–₹18,000 (depending on display, capacity)
- **Comparison with Our Solution:** Mantra's devices provide ready-to-use facial attendance with multimodal biometrics and local data logging, but lack QR-based GPS capture and deeper anti-spoofing checks like blink/reflection. Their recognition is embedded on device rather than in secure cloud models. Our solution adds QR location verification, encrypted facial encodings, and advanced spoof resistance with centralized database tracking.

6.8 Secureeye

- **Company Background & Establishment:** Secureeye is an Indian security and surveillance brand offering CCTV, access control, smart locks, and biometric attendance solutions across residential and business sectors. Known for "Make in India" products, it focuses on integrated security devices with nationwide support and cloud/web reporting options.
- **Implementation Approach:** Secureeye's face attendance devices capture live facial images via built-in cameras, then match them against stored user templates on device or cloud. They support multimodal verification (face, fingerprint, card, password) and push

data to attendance management software (Windows/Web/Cloud), providing LAN/Wi-Fi connectivity and basic anti-duplicate attendance protection.

- **Teck Stack:** On-device recognition firmware, Embedded Linux/RTOS (device OS), TCP/IP, Wi-Fi communication, Attendance management software integrations
- **Dataset used:** Proprietary enrolment face templates per organisation/device, No public benchmark dataset (custom per installation)
- **Product Specifications:**
 - Secureye S-FB6K Biometric Face & Fingerprint Attendance Machine: ~1000 face capacity, color TFT display, TCP/IP & Wi-Fi, cloud support.
 - Secureye Face attendance machine S-FB3K: ~300 face capacity, TCP/IP/Wi-Fi, LCD screen.
 - Secureye S-FB5K Elite Face Biometric Time & Attendance Machine: Mid-range face access device.
 - Screen sizes typically ~2.4"–3.5" on advanced models.
- **Market Price:**
 - S-FB6K: ~₹8,500–₹12,800.
 - S-FB3K: ~₹10,500.
 - S-FB5K: ~₹9,800.
 - Entry lineup around ~₹6,000–₹10,000 for basic face/finger devices.
- **Comparison with Our Solution:** Secureye's attendance devices offer embedded face and multimodal verification with onboard storage and cloud export, but lack QR-based GPS location capture and sophisticated anti-spoofing (blink/reflection) features. Their recognition runs on closed device firmware, not secure cloud encodings. Our solution adds location verification via QR, encrypted encodings, advanced liveness detection, and scalable database attendance tracking.

6.9 Salarybox

- **Company Background & Establishment:** SalaryBox is an Indian workforce management platform offering biometric attendance, HRMS, payroll, and cloud/mobile attendance solutions. It focuses on modernising attendance with AI face recognition, QR/GPS check-ins, and real-time dashboards for hybrid and field workforces. Headquartered in Gurugram, Haryana, it caters to thousands of Indian businesses.

- **Implementation Approach:** SalaryBox uses AI-powered face recognition devices and mobile apps to capture employee attendance via live face scans, QR codes, selfie/GPS check-ins, and cloud sync. Facial templates are stored securely, and devices push attendance to SalaryBox cloud/HRMS. It supports mask detection, mobile punch-ins, and real-time reporting with payroll integration.
- **Tech Stack:** AI face recognition algorithms (embedded/cloud), Wi-Fi, LAN, USB connectivity, Cloud attendance & mobile app integration, Database/cloud backend (AWS)
- **Dataset used:** Proprietary enrolled user face templates per organisation, Face data stored securely in cloud for recognition, Dynamic adaptation for lighting, masks, and angles.
- **Product Specifications:**
 - SalaryBox FaceScan Pro – AI face recognition with Wi-Fi, mask detection.
 - SalaryBox Gamma – Face recognition machine with 4.8" TFT screen and cloud connectivity.
 - Cloud/mobile attendance software Employee Smart Attendance, Live Tracking & Payroll Software – Web/mobile attendance/payslips integration.
 - Screen sizes: ~4.8" on advanced hardware models.
- **Market Price:**
 - FaceScan Pro: ~₹9,990.
 - Gamma: ~₹13,999.
 - Software suite: ~₹38,400.
 - Prices vary with features, cloud plans, and integrations
- **Comparison with Our Solution:** SalaryBox bundles AI face recognition hardware and mobile/cloud attendance with payroll, QR and GPS check-ins, and mask detection, but relies on proprietary cloud and prebuilt HRMS integrations. Our system adds secure facial encodings, deeper anti-spoofing (blink/reflection), QR-based GPS verification with proximity logic, and a centralized database tailored to academic attendance and proxy reduction.

6.10 ZKTeco

- **Company Background & Establishment:** ZKTeco is a global biometric and security tech company providing attendance, access control, and identity verification solutions across 100+ countries. Known for diversified biometric products (face, fingerprint, card),

the brand focuses on enterprise-grade workforce management, integrating attendance hardware with software tools and optional cloud/mobile interfaces.

- **Implementation Approach:** ZKTeco devices capture faces using visible-light or infrared cameras and run built-in algorithms like ZKFace to authenticate users against stored templates. They support multimodal recognition (face, fingerprint, card, password) and communicate via TCP/IP, Wi-Fi, or USB with on-device or cloud/PC attendance software, logging timestamps and exportable records for HR/payroll systems.
- **Tech Stack:** Embedded Linux/firmware, ZKFace recognition algorithms, TCP/IP, Wi-Fi communication, , On-device cameras & touch UI, Optional integration SDKs/attendance software
- **Dataset used:** Proprietary enrolled face templates stored on device, No public benchmark dataset — custom per deployment, Some models support large template capacities (up to ~6,000+)
- **Product Specifications:**
 - ZKTeco MiniTA Touchless Face Recognition Attendance & Access – Compact face attendance (~face-only units)
 - ZKTeco Touchless Face Multi-Biometric Time Attendance & Access – Face + fingerprint + card + battery backup
 - ZKTeco MB360: Face, Finger, and RFID Time Attendance & Access – Larger capacity multimodal unit
 - ZKTeco Face Attendance Machine with Access Control Functions – Mid-range face + RFID mix, Screen sizes typically 2.4"–4.3" touch/LCD depending on model.
- **Market Price:**
 - Entry-level facial units: ~₹7,000–₹9,000
 - Mid-range multimodal devices: ~₹12,000–₹18,000
 - Prices vary by reseller, model features, and capacities.
- **Comparison with Our Solution:** ZKTeco products offer ready-touse facial attendance with multimodal biometrics, local logs, and integration options, but they rely on preinstalled firmware recognition without dynamic QR/GPS check-ins or customized anti-spoofing (blink/reflection) logic. Our system adds secure facial encodings, QR-based

GPS verification, enhanced spoof checks, and centralized attendance storage tailored for campus proxy reduction and real-time analytics.

- Comparison Table:

Reference / Product	Limitations of Existing System	How Our System is Different & Better
eSSL	Closed firmware, no QR GPS verification, limited cloud flexibility	Software-driven system with QR GPS, encrypted encodings, cloud-based analytics
Mantra Softech	On-device recognition only, no QR GPS, limited spoof checks	Secure cloud encodings, QR GPS-based attendance, advanced liveness detection
Secureye	Basic anti-duplicate checks, no QR GPS validation	QR GPS verification, encrypted face data, stronger spoof resistance
SalaryBox	Proprietary ecosystem, limited academic customization	Academic-focused system, proximity-based QR GPS, deeper spoof detection
ZKTeco	Firmware-locked logic, no dynamic QR GPS verification	Customizable logic, QR GPS validation, encrypted facial encodings

Table 1: Market Products Comparison Table

7. Project Feasibility Study

7.1 Technical Feasibility

- The project uses existing and proven technologies such as Python, OpenCV, and face recognition libraries.
- Required hardware like a webcam and laptop is easily available.
- The system can run on standard computers without specialized hardware.

7.2 Economic Feasibility

- The project uses open-source software and libraries.
- Only basic hardware is required.

7.3 Operational Feasibility:

- The system is easy to use and requires minimal user interaction.

- Attendance is automatically recorded without manual effort.

7.4 Schedule Feasibility

- The project can be completed within the academic timeline.
- Each module can be developed and tested independently.

8. Project Requirement Gathering

8.1 Requirement Gathering Techniques Used:

- Literature Review: Studied existing face recognition and attendance systems. Reviewed online research papers and tutorials.
- Observation: Observed traditional attendance systems and their drawbacks.
- Online Research: Analyzed open-source face recognition projects and tools.
- User Feedback: Considered feedback from students and faculty regarding attendance issues.

8.2 Comparison of Existing Applications with Proposed Application:

Feature	Existing Systems	Proposed System
Attendance Method	Manual / Card-based	Face recognition
Time Consumption	High	Low
Accuracy	Prone to errors	High accuracy
Proxy Attendance	Possible	Reduced
Cost	Medium to High	Low
Automation	Partial	Fully automated
Anti-Spoofing	Not available	Basic spoof detection

Table 2: Comparison Study

9. Project Timeline Chart



Figure 1: Project Timeline Chart

10. Project Work Distribution Among Team Members

Students 23IT403, 23IT431, and 23IT435 collaboratively worked on the Face Recognition Attendance System. The team handled requirement gathering and system analysis, designed the overall workflow, and developed the face capture module. We organized and preprocessed datasets, performed face encoding and model training, and fine-tuned accuracy through optimization techniques. Anti-spoofing features were implemented to prevent false attendance. The system also includes camera handling with real-time location tracking, robust attendance database management, error handling, and validation. Detailed documentation was prepared, and the complete system was thoroughly tested and debugged to ensure reliable attendance report generation and smooth integration.

Guide Comments and Signature: